

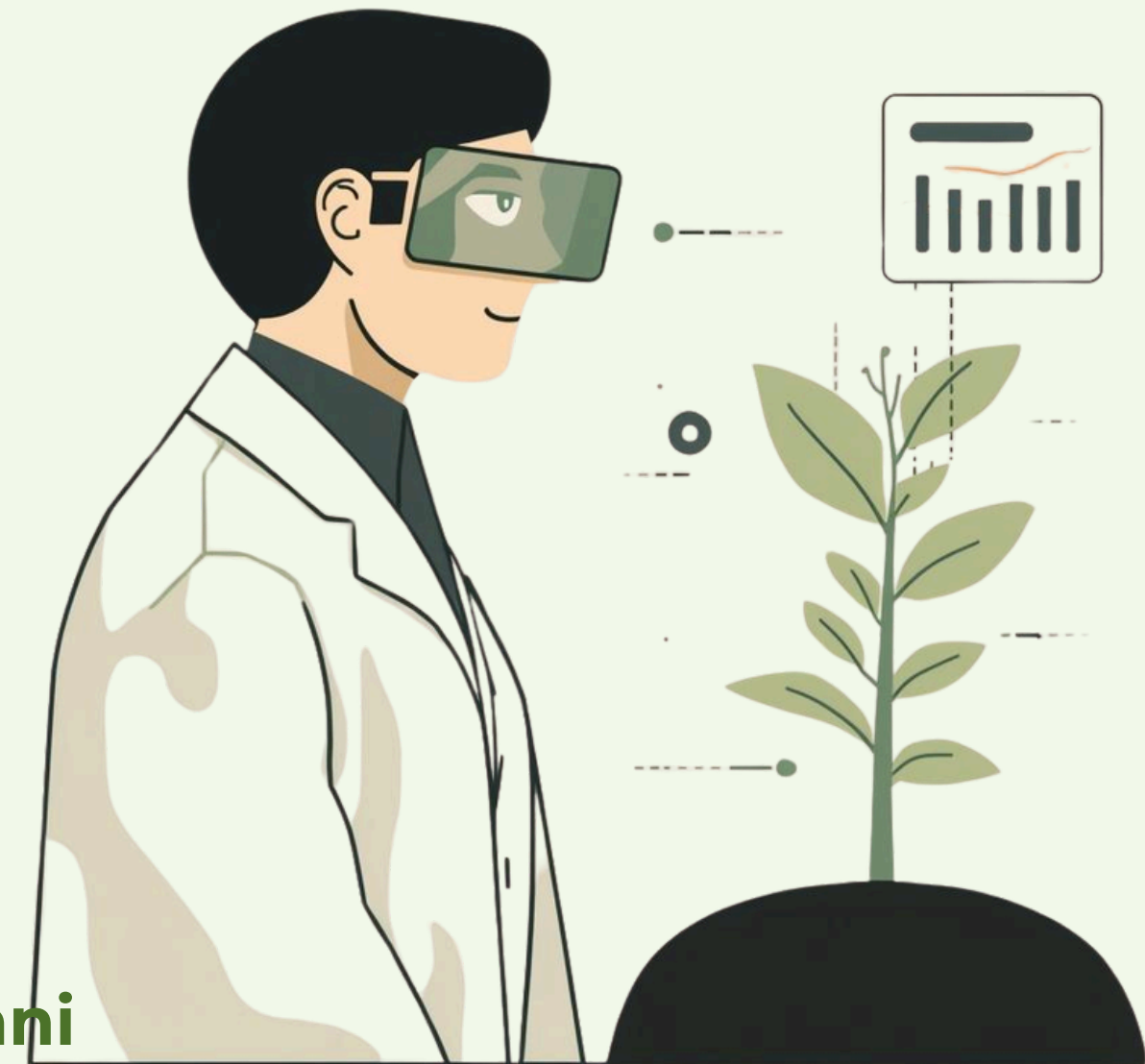
 AI Course Project

AgriSafe: Intelligent Farming Agent Under Noisy and Misleading Sensor Conditions

Smart agriculture under uncertain sensor data

Team Members

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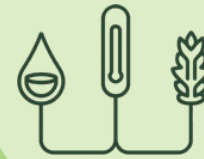
Introduction to AgriSafe

Project Scope & Objectives



Crop Health Focus

AgriSafe monitors crop conditions and supports healthier growth by helping farms respond early to changing plant needs.



Sensor-Based Decisions

The system uses soil moisture, temperature, and plant health readings to trigger automated actions and recommendations.



Why Smart Farming

AI-driven agriculture improves efficiency, supports consistent crop care, and helps farmers make better decisions from real-time data.

Problem Statement

Reliable Farming Under Sensor Noise



Noisy Sensor Data

Smart farming depends on soil moisture, temperature, and plant health sensors, but real readings can be noisy, delayed, or partially corrupted.



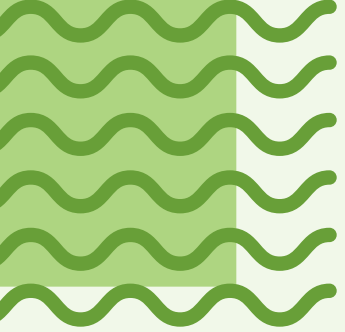
False Signals, Wrong Actions

When sensors report incorrect values, an AI agent may over-water crops, miss stress signals, or trigger alerts at the wrong time.



Adversarial Uncertainty

The core challenge is maintaining plant health in an environment where sensors can fail, drift, or provide misleading information.



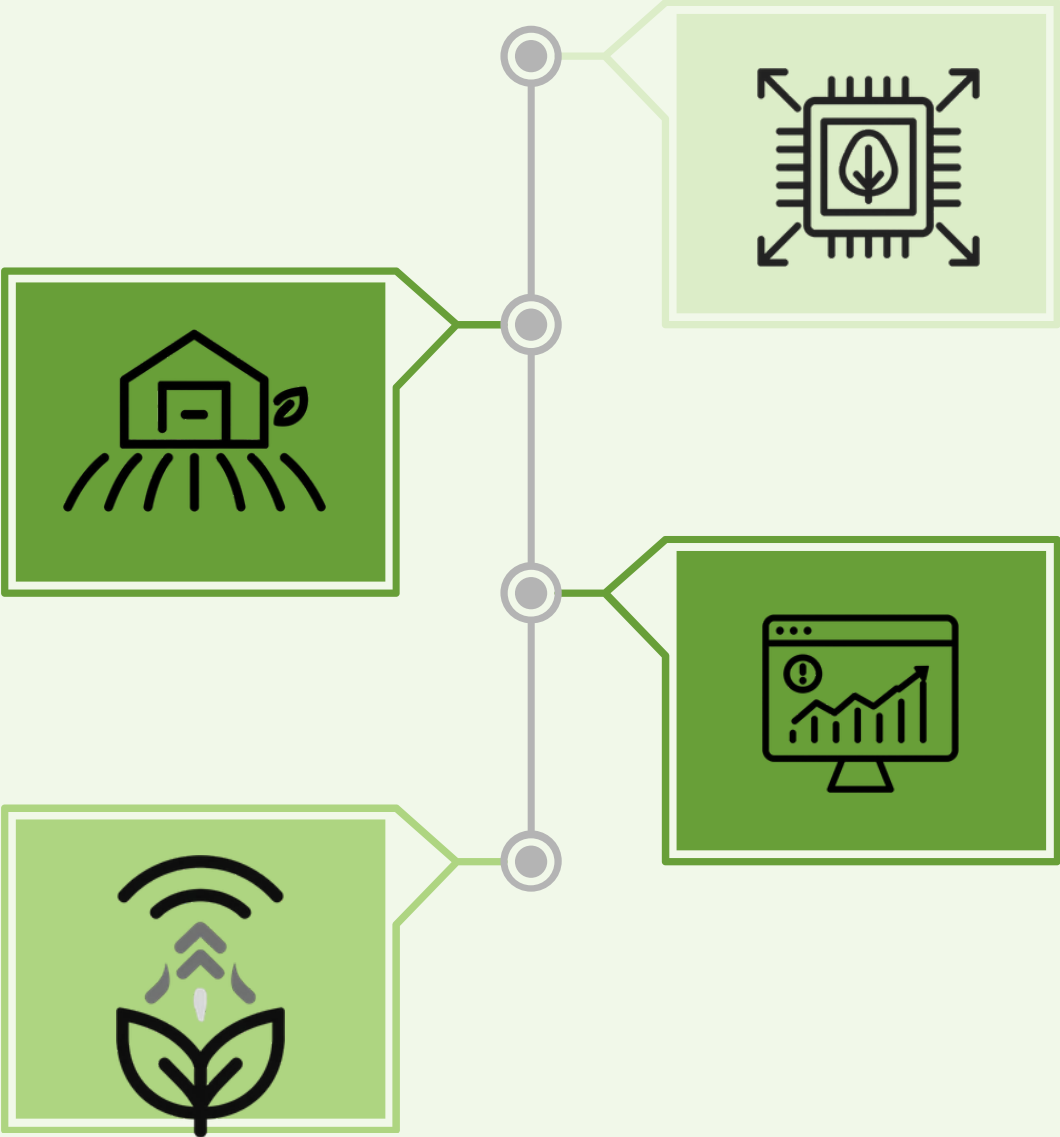
System Design Overview

Simulated Farm Environment

A virtual crop field models soil moisture, temperature, and plant health so the agent can operate in controlled farming scenarios.

Sensor System

Soil moisture, temperature, and plant health sensors provide the observations used to monitor field conditions and detect changes.

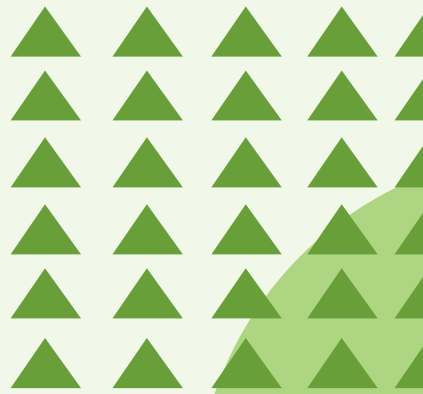


Rule-Based AI Agent

The agent applies predefined if-then rules to interpret sensor readings and decide when to water crops or trigger alerts.

Evaluation System

Performance is compared under normal and noisy sensor conditions to measure robustness, decision quality, and plant health stability.



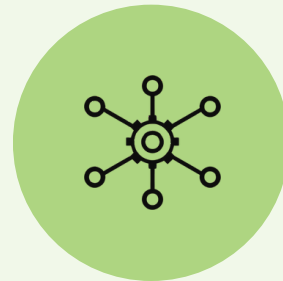
AI Agent Logic

Perceive, Decide, Act



Perception: Read Sensors

The agent reads soil moisture, temperature, and plant health signals from the environment.



Decision: Apply Rules

It applies if-then rules to interpret readings and choose the safest next step.



Action: Water or Alert

The system waters crops or triggers alerts to protect plant health in real time.

Adversarial Conditions in Sensors



Random Noise

Small fluctuations are added to moisture, temperature, and plant-health signals to mimic unstable sensing and obscure the real crop state.



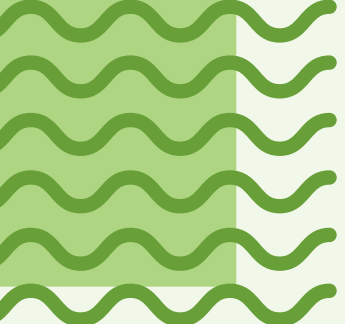
Sensor Bias

Some readings are shifted consistently too high or too low, simulating calibration drift that can mislead the agent's decisions.



Faulty Readings

Dropped, spiking, or conflicting values simulate sensor faults and test whether the agent remains reliable under uncertainty.



Plant Health Stability Under Sensor Noise



Stable in Normal Conditions

With accurate readings, the agent kept plant health steady through timely watering and heat alerts.



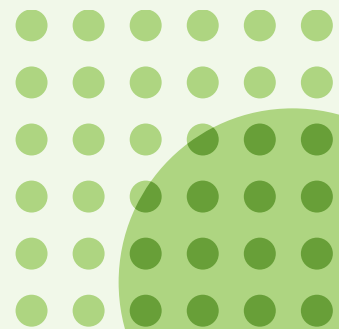
Resilient Under Noise

Even with misleading moisture and temperature data, plant health remained largely stable across the simulation.



Small Performance Gap

Results showed only a modest drop versus normal conditions, highlighting robust decisions under uncertainty.



Decision Accuracy & Reliability



Correct Watering Decisions

The agent waters crops only when low moisture signals align with plant needs, limiting unnecessary irrigation under noisy readings.



Reliable Alert Triggering

Temperature and plant-health alerts stay timely and consistent, even when misleading sensor values appear in the environment.



Reduced False Decisions

False positives and false negatives are reduced, improving trust in the system's day-to-day recommendations.

Conclusion: Robust AI for Smart Farming



Reliable Under Noisy Inputs

AgriSafe makes consistent watering and alert decisions even when sensor readings are noisy or misleading.



Robust for Smart Farming

Stable plant health in tough conditions shows clear potential for real-world smart farming deployment.



Value Under Uncertainty

The project proves simple rule-based AI can create value when agriculture systems face uncertainty.

Future Work

Add Machine Learning

Train adaptive models to update rules over time and improve responses to changing farm conditions.

Expand Sensors and Data Fusion

Combine soil, temperature, humidity, and plant signals so decisions rely on richer and more reliable inputs.

Field Deployment and Stress Tests

Evaluate AgriSafe in physical farming setups and under tougher adversarial cases such as biased or faulty readings.



Thank You!

Questions are welcome – let us discuss AgriSafe

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